# The Legendre Transform

The **Legendre transform** is a method to express the information encoded in a given function in a different way. Instead of describing a function by its original variable, we describe it using its **slope**.

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## 1. Original Function

Let

$$L = L(V). \tag{1}$$

Define

$$P(V) = \frac{\partial L(V)}{\partial V}. \tag{2}$$

Here, $P$ is the slope function of $L(V)$.

Equivalently,

$$L(V) = \int_0^V P(V')\, dV'. \tag{3}$$

So $L(V)$ can be interpreted as the area under the curve $P(V)$.

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## 2. Rotating and Flipping the Picture

The Legendre transform changes variables from $V$ to $P$.

If

$$P = \frac{\partial L}{\partial V}, \tag{4}$$

then, assuming this relation can be inverted, we can write

$$V = V(P). \tag{5}$$

The new function is

$$H(P) = \int_0^P V(P')\, dP'. \tag{6}$$

This function depends on the slope variable $P$, rather than the original variable $V$.

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## 3. Relation Between $L(V)$ and $H(P)$

The areas satisfy

$$L(V) + H(P) = PV. \tag{7}$$

Therefore,

$$H(P) = PV - L(V). \tag{8}$$

This is the Legendre transform of $L(V)$.

In words:

> We switch from a function $L(V)$, which depends on $V$, to a new function $H(P)$, which depends on the value of the slope $P$ at each point.

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# 4. Mechanical Application

In mechanics, the generalized momentum is defined by

$$p = \frac{\partial L}{\partial \dot{q}}. \tag{9}$$

The Hamiltonian is the Legendre transform of the Lagrangian with respect to $\dot{q}$:

$$H(q, p, t) = p\dot{q} - L\Big(q, \dot{q}(q, p), t\Big). \tag{10}$$

Here $\dot{q}$ is eliminated in favor of $p$.

From the Hamiltonian, one obtains

$$\dot{q} = \frac{\partial H}{\partial p}. \tag{11}$$

Thus the Legendre transform changes the description from the variables

$$(q, \dot{q}) \tag{12}$$

to the variables

$$(q, p). \tag{13}$$